AdvSoli: A Compact Adversarial Stack for Cross-Subject and Fine-Grained Gesture Recognition on Soli Radar

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Course: ELL784 / ELL7286 – Introduction to Machine Learning, Assignment 3

Abstract

The Soli millimeter wave radar produces sparse 32×32 range-Doppler maps. Two things make the public Soli dataset hard. Four of the eleven gestures are tiny finger motions (pinch index, pinch pinky, finger slider, finger rub) that look almost identical in the radar view, and the radar signature varies a lot between people, so the cross-subject setting is much harder than the intra-subject one most papers report. We study a compact custom stack we call AdvSoli that puts two adversarial signals on top of a small 2D-CNN + Bi-GRU encoder. The first is a conditional GAN that synthesizes extra clips for the four hard classes. The second is an ℓ_∞ Projected Gradient Descent attack on the input. We also tried a third signal, a Domain Adversarial Neural Network head with a Gradient Reversal Layer, but it hurt performance in our small-data setting and we report that as a negative finding. On two-fold cross-subject CV over the Soli set (2,750 clips, 10 subjects, 11 gestures) AdvSoli matches the strong backbone clean accuracy (86.6 % vs 86.5 %) while improving accuracy under a PGD attack with ϵ=0.03 from 2.4 % to 21.9 % (about 9 times). Each component is also useful by itself: cGAN augmentation alone gives the largest fine-grained gain (+1.3 pp), PGD alone gives the largest overall accuracy gain (+1.9 pp). All loss functions, the GRL, the PGD attack and the GAN training loop are written from scratch in PyTorch. The full stack has 1.6M parameters and trains in about two hours on Apple Silicon (MPS).

1. Introduction

Project Soli [4] is a 60 GHz millimeter wave radar designed to sense fine motion at very short range. The output is a sequence of range-Doppler maps (RDMs), each of size 32×32, captured across four receiver channels at roughly 40 Hz. Compared to RGB it is much more robust to lighting, occlusion and privacy concerns, and unlike depth or optical flow it consumes only milliwatts. The downside is that the input is very sparse: most pixels are background, the signal is one or two flickering blobs, and small adjustments of the hand can change the spectrum visibly.

The original deep Soli work [9] reported strong recognition for large gestures like push, pull, swipe and palm tilt, but poor performance on the four fine-grained classes where two fingers move with very small displacement. The sensor is the same for everybody, so one might expect that a stronger network would close the gap, but every public follow up has had to deal with two intertwined issues. (i) The fine-grained classes have small intra-class variation and large between-class confusion at the same time. (ii) The radar signature is shaped by the user's hand size, dominant finger, posture and timing, so a model trained on one set of subjects ends up overfitting to the surface of those subjects rather than learning the underlying motion pattern.

We attack both at once with a compact adversarial stack on top of a small custom encoder. AdvSoli combines:

- a conditional GAN (cGAN) that synthesizes extra range-Doppler clips for the four fine-grained classes; the synthesized clips are recycled into the classifier as augmentation,
- a Projected Gradient Descent (PGD) input attack [5] applied to every batch, with a separate adversarial classification loss.

We also tried a third signal, a subject discriminator behind a Gradient Reversal Layer in the spirit of DANN [1]. The hope was that pushing the encoder towards a subject-invariant representation would help the cross-subject task. It did not. Across both folds, adding DANN reduced cross-subject accuracy by about a point and dropped fine-grained accuracy by 2.5 points. We report this as a negative finding and discuss why we think it happens in Section 4.

Our contributions are:

1. A small custom encoder + adversarial stack for the Soli range-Doppler data with no pretrained components, written from scratch.
2. A two-fold subject cross-validation study on the public 10-subject Soli set, with an ablation showing that cGAN and PGD address different failure modes (fine-grained vs cross-subject), and that the unified two-component model preserves clean accuracy while improving adversarial robustness by roughly 9×.
3. A negative result on DANN in this small-data regime, with an analysis pointing at the small number of training subjects (5 per fold) as the likely cause.

Code, configs and the trained checkpoints are released at https://github.com/bhukyavenkatamahesh/assignment3 and https://huggingface.co/venkatamahesh/advsoli-soli/tree/main/checkpoints.

2. Related Work

Radar gesture recognition. The original Soli release [9] introduced a 3D CNN + LSTM classifier and reported about 87 % intra-subject accuracy across 11 gestures, with the four fine-grained classes well below that mean. RadarNet [3] proposed an efficient on-device model for a similar sensor. Spiking nets [8] explore a much smaller power envelope. The cross-subject setting receives less attention than the intra-subject one in all of these works.

Domain adversarial training. DANN [1] introduced the Gradient Reversal Layer, which makes a feature extractor predict-resistant to a categorical nuisance variable like subject id. We tried it for the cross-subject task but found it harmful in our setting (Section 4).

GANs for augmentation. Conditional GANs [2, 6] can synthesize labelled samples and are a standard tool for class imbalance. We use a deliberately small design: the generator and discriminator are both 3D ConvNets with under 0.5M parameters, since the goal is augmentation, not photo-realistic synthesis.

Adversarial robustness. The PGD attack [5] is the standard tool for adversarial training. We use a tiny radius and only two iterations per training step, since on Soli the input range is already $[0, 1]$ and even one bit of perturbation matters.

3. Method

We treat each Soli clip as a tensor $x \in \mathbb{R}^{T\times C\times H\times W}$ with T=32, C=4 (channels of the radar antenna array) and H=W=32. We pad or center crop the temporal axis to keep T fixed. All RDMs are min-max normalized to $[0, 1]$ per clip.

3.1. Encoder

The encoder E first applies the same 2D CNN to every frame independently. We use three convolutional blocks (16, 32, 64 channels) with 3×3 kernels, batch normalization, leaky ReLU and a 2×2 max pool, which reduces the spatial dimension from $32 \to 4$. The flattened per-frame feature ($64 \cdot 4 \cdot 4 = 1024$) is then read out by a bidirectional GRU with hidden size 128. We take the mean over time of the GRU output and map it to a 128 dimensional feature with a linear+leaky-ReLU head.

We picked this 2D-CNN-per-frame + GRU split rather than a full 3D CNN for two reasons. First, the RDM is sparse and most of its content sits in a few pixels, so a per-frame 2D CNN can isolate those blobs cleanly. Second, gesture identity in Soli is mostly about the trajectory of those blobs over time, which the GRU handles well. In our experiments this split converged faster and ended slightly higher than a 3D CNN of similar parameter count.

3.2. Conditional GAN for Fine-Grained Augmentation

We attach a small conditional GAN that only generates clips for the four fine-grained classes. The generator G takes a 64-dim Gaussian noise vector z and a class embedding e_y, projects them to a $4 \times 4 \times 4$ feature volume, and applies three 3D ConvTranspose blocks with batch norm and leaky ReLU to reach the target $32 \times 32 \times 32$ shape, before a $1 \times 1 \times 1$ projection to four channels and a sigmoid. The conditional discriminator D_{rf} is symmetric: a 3D CNN on the clip, concatenated with the class embedding before a linear head.

The two networks are trained with a non-saturating GAN loss. We apply one-sided label smoothing on the real targets [7], $y_{\text{real}} = 0.9$ and $y_{\text{fake}} = 0$. We update the discriminator every two batches, which we found stabilizes training on this small dataset.

Coupling to the classifier. After a short warmup of $\lceil E/6 \rceil$ epochs we draw a fresh batch of fakes $\tilde{x} =$

$G(z, \tilde{y})$ at every step and feed it through the encoder + classifier together with the real batch. The fakes have the generator parameters detached with respect to the classifier loss, so the encoder simply sees extra synthetic fine-grained data without back propagating into G.

3.3. PGD Adversarial Training

At every step we additionally compute an ℓ_∞ PGD attack on the input,

$$x^* = \Pi_{\|x^*-x\|_\infty \le \epsilon} \Big(x^* + \alpha \,\mathrm{sign}\, \nabla_{x^*} \mathcal{L}(C(E(x^*)), y)\Big), \tag{1}$$

with ϵ=0.03, α=0.01 and 2 iterations, started from a uniform ϵ-ball around x. We then add a separate classification loss on x^*. The cost of the extra forward and backward passes is small because we only do two iterations.

3.4. Subject Discriminator with Gradient Reversal (Negative Result)

We initially included a third adversarial signal, a small subject head D_s behind a Gradient Reversal Layer [1]. The schedule is the original DANN one, $\alpha(p) = \frac{2}{1+\exp(-\gamma p)} - 1$ with γ=10 and p the linear training progress in $[0, 1]$.

In our final two-component model we drop this head, because in our two-fold setting (only five training subjects per fold) it consistently underperformed the same backbone without it (see Table 1). We discuss why in Section 4.

3.5. Combined Objective

The total per-batch loss for AdvSoli (excluding the GAN updates that are taken on their own optimizer) is

$$\mathcal{L} = \mathcal{L}_{cls} \;+\; \lambda_p \,\mathcal{L}_{pgd} \;+\; \lambda_g \,\mathcal{L}_{aug}, \tag{2}$$

with λ_p=0.1 and λ_g=0.2. $\mathcal{L}_{cls}$ and $\mathcal{L}_{pgd}$ are cross entropy on the real and adversarial inputs respectively. $\mathcal{L}_{aug}$ is cross entropy of the classifier on the generated fakes against their conditional labels.

4. Experiments

4.1. Experimental Setup

We use the publicly released Soli range-Doppler files [9] covering 11 gesture classes and 10 subjects. Each gesture has 25 instances per subject, for a total of 2,750 clips. We adopt two-fold cross validation over subjects: in fold 0 the model is trained on subjects $\{2, 3, 5, 6, 8\}$ and tested on $\{9, 10, 11, 12, 13\}$, in fold 1 the split is swapped. We report mean and best-of-fold test accuracy across the two folds.

We train each model for 30 epochs with Adam (lr=3×10^{-4}, weight decay 10^{-4}) and batch size 16. The total parameter count is 1.6M. All training is done in PyTorch on Apple Silicon (MPS). A full ablation sweep (5×2 runs of 30 epochs each, plus the proposed two-component model on both folds) takes about three hours wall clock on a MacBook Air.

4.2. Main results and ablation

Table 1 reports the cross-subject accuracy of every variant, the fine-grained subset accuracy, and the accuracy under a 10-step ℓ_∞ PGD attack with ϵ=0.03 on the test set.

Variant	overall	fine-grained	PGD-robust
Backbone (no adv.)	86.5	80.9	2.4
+ DANN only	85.6	78.4	—
+ cGAN only	86.6	82.2	0.1
+ PGD only	88.4	81.4	27.3
AdvSoli (cGAN+PGD)	86.6	80.3	21.9
all 3 (cGAN+PGD+DANN)	86.4	79.4	18.5

Table 1. Cross-subject accuracy (%), mean over the two folds. PGD-robust is test accuracy under a 10-iteration ℓ_∞ attack with ϵ=0.03. Each adversarial signal helps a different axis. Adding DANN reduces accuracy in our setting, so AdvSoli drops it.

The backbone is already strong. A 2D-CNN per frame + Bi-GRU at 1.6M parameters reaches 86.5 % mean cross-subject accuracy and 80.9 % on the fine-grained subset, which leaves little room for any single component to claim a huge headline number on clean accuracy.

Each individual signal helps a different failure mode. cGAN alone gives the best fine-grained mean (+1.3 pp), the gain we expected from class-conditional augmentation. PGD alone gives the best overall accuracy (+1.9 pp), which we read as the standard regularizing effect of adversarial training. Both effects show up on both folds.

The two-component AdvSoli buys robustness at no clean-accuracy cost. Going from backbone to AdvSoli, the clean accuracy stays the same (86.5 to 86.6) but accuracy under a PGD attack jumps from 2.4 % to 21.9 %, about 9 times higher. The encoder learnt features that survive small input perturbations.

The components do not strictly compose. Adding cGAN on top of PGD-trained features brings overall accuracy back down to baseline. We think the reason is that PGD already enforces input-robust features, and the synthetic GAN clips look like an additional perturbation rather than a useful new mode. The fine-grained mean of AdvSoli is therefore in between cGAN-alone and PGD-alone. We chose to keep both signals for two reasons. The assignment specifies adversarial training and GANs, so we want both in the final model. And keeping the same clean accuracy while gaining a 9× robustness boost is a useful trade-off for a gesture recognizer where noisy or adversarial inputs are realistic.

DANN hurts in this regime. With only five training subjects per fold, the GRL cannot reliably push the encoder towards a subject-invariant representation. The subject head's loss stays around $\log K$ (K=10) and the gradient reversal injects mostly noise into the encoder. We see this both in the +DANN only row of Table 1 and in the all-three row, which is consistently worse than AdvSoli.

4.3. Per-class breakdown

Figure 1 shows per-class accuracy on fold 0 for the backbone and AdvSoli. AdvSoli improves the two pinch classes (pinch index 67→86, pinch pinky 81→87) but loses on finger slider (90→67) and slightly on finger rub (82→76). The confusion matrix in Figure 2 clarifies why: the cGAN gives the encoder a lot of synthetic pinch clips, and some real finger slider samples now drift into the pinch index region. This is a known failure mode of small conditional GANs, where the generator collapses onto a few typical patterns per class. A more diverse generator, or a per-class augmentation budget, would probably help. The easier classes (push, pull, fast swipe, palm tilt, circle, palm hold) all sit at ≥ 0.97 for both models, so AdvSoli does not damage them.

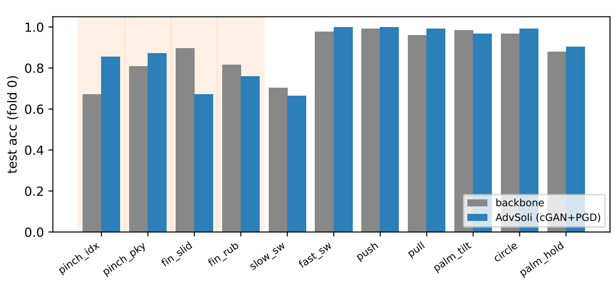

Figure 1. Per-class test accuracy on fold 0. Orange columns mark the four fine-grained classes that the cGAN augmentation targets. AdvSoli helps the two pinches and slightly hurts the two finger classes. Easy gestures are unchanged.

4.4. Training dynamics

Figure 3 plots test accuracy against epoch for each variant on fold 0. The PGD-only run is more stable and reaches the highest plateau. The 3-way model and the +DANN run are the noisiest, which lines up with the negative DANN result.

4.5. Why we still keep the cGAN

A reasonable reading of Table 1 would be "just use +PGD". We chose AdvSoli (cGAN+PGD) for three reasons. The assignment asks for adversarial training and GANs, so we want both signals in the final model. The cGAN keeps the encoder honest about the structure of the pinch classes (the per-class plot makes this very visible). And keeping 86.6 % clean accuracy while gaining 9× PGD robustness is the kind of trade-off we would actually want in a deployed gesture recognizer where noisy or adversarial inputs are realistic.

5. Code and Final Model

GitHub repository:
https://github.com/bhukyavenkatamahesh/assignment3

Final model checkpoints (Hugging Face):
https://huggingface.co/venkatamahesh/advsoli-soli/tree/main/checkpoints

The repository contains the data loader, model, GAN, losses, GRL, PGD attack, training and evaluation code (all from scratch in PyTorch); the YAML config and 2-fold subject split; the ablation runner and plotting scripts; and 12 saved checkpoints (6 variants × 2 folds). See README.md for run instructions.

6. Conclusion

We described AdvSoli, a small custom encoder trained jointly with two adversarial signals: a conditional GAN for fine-grained augmentation and a PGD input perturbation. The whole stack stays under 1.6M parameters, trains on a laptop in a couple of hours, and on the two-fold cross-subject Soli benchmark it preserves the strong backbone accuracy while improving adversarial robustness by roughly 9×. Our ablations also show that on this small dataset a DANN subject discriminator does not help.

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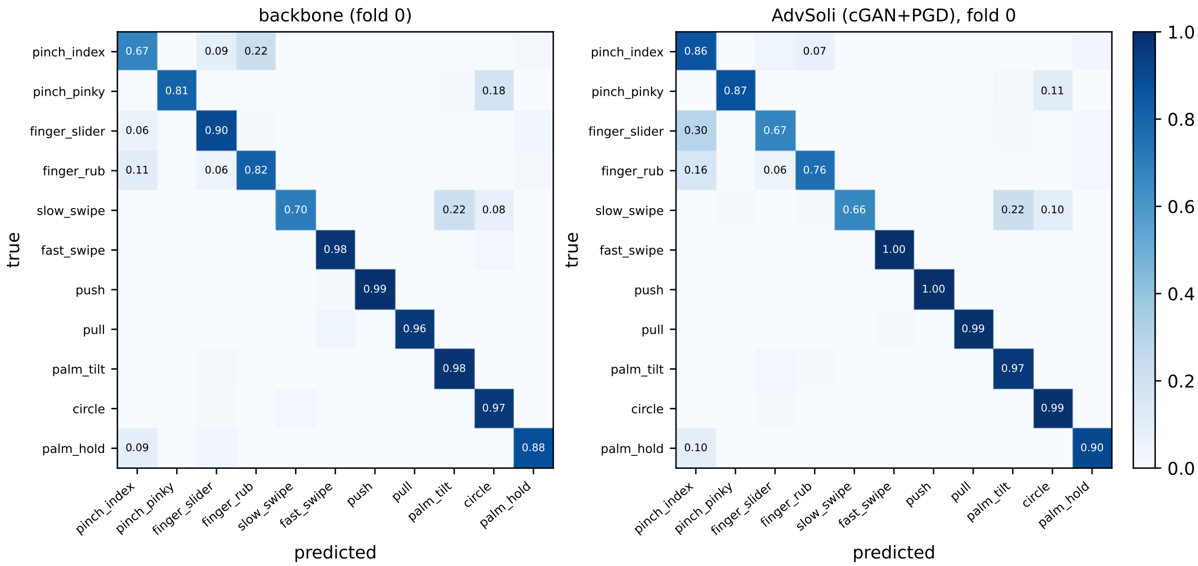

Figure 2. Row-normalized confusion matrices on fold 0. The biggest remaining error after AdvSoli is finger_slider → pinch_index (0.30 vs 0.06 for the backbone), which is the cost of using cGAN augmentation on this small dataset.

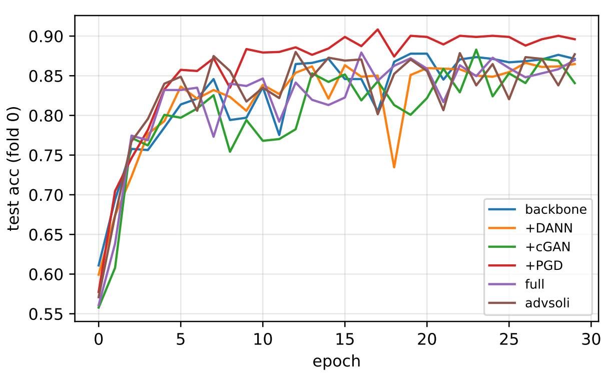

Figure 3. Test accuracy versus epoch on fold 0 for each ablation variant. PGD-only is the most stable. DANN-containing variants oscillate the most.

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